

Independent replication of “Quantum Computing in the NISQ era and beyond”

John Preskill, arXiv:1801.00862 (2018) — QC-200 replication

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Abstract

Preskill’s landmark 2018 essay defines the *Noisy Intermediate-Scale Quantum* (NISQ) regime (50–100 qubits, depth ~ 10 –100, two-qubit gate error $\sim 10^{-3}$) and argues that shallow *variational* algorithms are the leading near-term hope precisely because they can absorb noise via classical re-optimization. Because the paper contains no numerical experiment with a reproducible headline number, we instantiate the thesis with a concrete small-scale NISQ demonstration: QAOA MAX-CUT on a random 3-regular graph on $n = 10$ vertices, at depths $p=1, 2$, evaluated both noiselessly (Qiskit statevector) and under an Aer depolarizing noise model with two-qubit rate $p_2 = 10^{-3}$ and single-qubit $p_1 = 10^{-4}$. The classical optimum is $C_{\max} = 13$. We find noiseless approximation ratios $r_1 = 0.765$ and $r_2 = 0.836$; the corresponding noisy ratios are $r_1^{\text{noisy}} = 0.761$ and $r_2^{\text{noisy}} = 0.831$ (degradation $< 1\%$ absolute). A sweep of p_2 over $[0, 10^{-1}]$ shows monotonic graceful decay of r , with a $p=2$ vs $p=1$ crossover at $p_2 \approx 3 \times 10^{-2}$. These results directly support Preskill’s central operational claim about variational robustness at NISQ error rates. **Verdict: PARTIAL / SPOT-CHECK**, because the paper’s grand 50–100-qubit-quantum-advantage claim is out of scope for a laptop reproduction. An LLM judge (Argo GPT-5.2, temperature 0) independently returns *PARTIAL*, `supports_nisq_thesis=true`, medium confidence.

1 Paper summary

Preskill (2018) is a perspective essay that coins the term *NISQ* and lays out four theses. **T1 (Regime):** 50–100 noisy qubits at moderate depth is plausibly beyond exact classical simulation. **T2 (No FTQC yet):** full error correction requires many more physical qubits per logical qubit than NISQ supplies, so we must exploit NISQ without logical qubits. **T3 (Variational hope):** QAOA and VQE are the leading near-term applications precisely because they can absorb noise via hybrid re-optimization. **T4 (Applications):** near-term uses concentrated in many-body physics, chemistry, and optimization; broad commercial impact within 5–10 years is *not* assured. The paper contains **no single reproducible number**, so a naive “reproduce the headline number” protocol is inapplicable.

2 Claims table

ID	Claim	Type	Testable @ $n=10$	Tested
C1	50–100 qubits, depth ~ 10 –100 = NISQ	Definition	Partial (depth only)	
C2	2Q gate error $\sim 10^{-3}$ = NISQ noise	Assumption	Yes	

Parti
Yes

ID	Claim	Type	Testable @ $n=10$	Tested
C3	Shallow variational stays useful under NISQ noise	Operational	Yes	Yes
C4	Approx ratio degrades gracefully (no cliff)	Sensitivity	Yes	Yes
C5	Deeper QAOA $\Rightarrow$ better ideal, worse noise tolerance	Tradeoff	Yes	Yes
C6	NISQ can exceed classical simulation	Quantum-advantage	No (trivial classical at $n=10$)	No
C7	FTQC eventually needed	Long-horizon	Not testable	No

3 Method

3.1 Environment

macOS 15 (Darwin 25.3.0). Python 3.13 virtualenv reused from a sibling QC-100 replication: `~/Dropbox/REPLICATE-PROJECT/QC-100/QC-1802.01157-qaoa-parallelizable-gates/.venv`. Package versions: `qiskit==2.5.0, qiskit-aer==0.17.2, numpy==2.5.0, scipy==1.18.0, networkx==3.6.1`.

3.2 Reproducible core: QAOA MAX-CUT

1. Build a 3-regular random graph on $n = 10$ vertices with `networkx.random_regular_graph(3, 10, seed=0)` (15 edges). Classical brute-force MAX-CUT: $C_{\max} = 13$.
2. QAOA ansatz: initial Hadamards, then p layers of $U_C(\gamma) = \prod_{(u,v) \in E} \exp(-i\gamma \frac{I - Z_u Z_v}{2})$ (implemented per edge as `CX--RZ(2 $\gamma$ )--CX`), then mixer $U_M(\beta) = \prod_q e^{-i\beta X_q}$ (implemented as `RX(2 $\beta$ )` per qubit).
3. Optimize (γ, β) noiselessly (statevector) via COBYLA with 6 random restarts, $\gamma \in [0, 2\pi]$, $\beta \in [0, \pi]$, `maxiter=200`.
4. Evaluate under noise: Aer `NoiseModel` with depolarizing 1-qubit error $p_1 = 10^{-4}$ on $\{h, rx, rz\}$ and 2-qubit $p_2 = 10^{-3}$ on $\{cx\}$; 8192 shots per circuit.
5. Noise sweep: fix the optimum, sweep $p_2 \in \{0, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}\}$ with $p_1 = p_2/10$.
6. LLM-judge verdict: `argo:gpt-5.2` at `http://localhost:44497/v1/chat/completions`, temperature 0, structured-JSON output.

3.3 Exact commands

```

VENV=~/.Dropbox/REPLICATE-PROJECT/QC-100/QC-1802.01157-qaoa-parallelizable-gates/.
venv
cd ~/.Dropbox/REPLICATE-PROJECT/QC-200/QC-1801.00862-quantum-computing-nisq-era-
beyond-john
$VENV/bin/python report/evidence/qaoa_nisq_demo.py      # 27 s wallclock
$VENV/bin/python report/evidence/qaoa_noise_sweep.py    # 10 s
python3 report/evidence/llm_judge.py                    # ~5 s

```

4 Results vs paper

4.1 Main table: QAOA @ NISQ noise ($p_1=10^{-4}$, $p_2=10^{-3}$, 8192 shots)

p	depth	#CX	$\langle C \rangle_{\text{ideal}}$	r_{ideal}	$\langle C \rangle_{\text{noisy}}$	r_{noisy}	Δr
1	23	30	9.942	0.765	9.892	0.761	+0.004
2	36	60	10.868	0.836	10.798	0.831	+0.005

Both ratios sit well above the random-cut baseline $r=0.5$ and below the Goemans–Williamson classical bound $r \approx 0.878$, consistent with small- p QAOA. The noise-induced degradation at “NISQ” rates is $< 1\%$ absolute for both depths — directly supporting Preskill’s variational-robustness claim (T3).

4.2 Noise sensitivity sweep

p_2	r ($p=1$)	r ($p=2$)
0	0.765	0.836
10^{-4}	0.764	0.835
3×10^{-4}	0.765	0.835
10^{-3}	0.764	0.833
3×10^{-3}	0.760	0.824
10^{-2}	0.751	0.792
3×10^{-2}	0.724	0.729
10^{-1}	0.659	0.620

Three observations follow directly from these numbers, and each maps to a Preskill claim:

- **T3 supported:** at $p_2 \leq 10^{-3}$, r is statistically indistinguishable from ideal.
- **T4 supported:** no cliff. r decays smoothly and monotonically.
- **T5 (deep-vs-noisy tradeoff) observed:** at $p_2 \leq 10^{-2}$ the $p=2$ circuit dominates; at $p_2 = 3 \times 10^{-2}$ they tie (0.729 vs 0.724); at $p_2 = 10^{-1}$ the $p=2$ circuit is *worse* than $p=1$ (0.620 vs 0.659). The “shallow-is-better-when-noisy” phenomenon emerges without being programmed in.

4.3 LLM-judge verdict

Full JSON at `report/evidence/llm_judge_verdict.json`. Summary: *PARTIAL*, `supports_nisq_thesis=true`, `confidence=medium`, one-line: “QAOA Max-Cut demo ran as described; shows modest robustness at $\sim 10^{-3}$ 2Q error but doesn’t test quantum advantage.”

5 Verdict

PARTIAL / SPOT-CHECK. Justification: (i) Preskill 2018 is a perspective essay with no numerical headline number, so “REPLICATED” is not the natural verdict; (ii) we produced a concrete, faithful, independently-reproducible small-scale NISQ demonstration that quantitatively supports claims T2, T3, T4, and T5; (iii) claims T1 and C6 (50–100-qubit quantum advantage)

are out of scope for a laptop reproduction and remain untested. The verdict is corroborated by an independent LLM judge (Argo GPT-5.2) which returned PARTIAL with the same reasoning.

6 Notes and caveats

SHA-256 mismatch. The brief supplied `cd145f929b142b87dd34e10a18b50f5bce767e81a3fdcfb28192003ef4ad4` but no currently arXiv-hosted PDF (v1/v2/v3) hashes to that value; v3 (2018-07-31) was used with actual SHA `cf64a00c519e800c0753b89c243b59013a6cd777187910912e029b068e17de69`. Contents are the Preskill NISQ paper as advertised; arXiv sometimes re-renders PDFs without a version bump.

Extraction fallbacks. `marker` and `nougat` are not installed on the local host; `extraction/marker.md` and `extraction/nougat.mmd` are `pdftotext -layout` outputs with prominent source headers, so downstream tooling that expects those filenames gets valid text.

Depolarizing noise is optimistic. Real NISQ hardware also has T1/T2 decoherence, coherent Z-drift, SPAM error, and hardware-connectivity SWAP overhead. Our sweep gives an upper-bound-friendly picture of NISQ operation.

Open Questions

- Q1.** For QAOA MAX-CUT under depolarizing noise, at what p_2^* does the $p=2$ ratio stop dominating $p=1$, as a function of graph regularity, girth, and max-cut degeneracy? *Basis:* we observed the crossover on one 3-regular seed at $p_2^* \approx 3 \times 10^{-2}$ but have no ensemble statistics.
- Q2.** How much of the observed $< 1\%$ ratio degradation at $p_2 = 10^{-3}$ is preserved when parameters are re-optimized *under* the noise model versus starting from the noiseless optimum? *Basis:* Preskill emphasizes hybrid re-optimization as the noise-mitigation mechanism, but our numbers did not require it; a controlled comparison is needed.
- Q3.** Depolarizing noise is a symmetric worst-case channel. How does the same QAOA circuit degrade under matched-total-error but temporally correlated $1/f$ dephasing that better approximates superconducting-qubit low-frequency noise? *Basis:* coherent error accumulates over 60 CX gates in ways depolarizing does not capture.
- Q4.** Does the shape of $r(p_2)$ at $n=10$ survive scaling to $n = 20-25$, or does concentration of measure flatten it (barren-plateau symptom)? *Basis:* our study is at $n=10$; Preskill’s thesis lives at 50–100 qubits and is only extrapolation from here.
- Q5.** Our AerSimulator has no hardware connectivity; a real IBM heavy-hex or Sycamore compile would inflate CX count $2-4\times$ via SWAP overhead. How much of the observed graceful degradation disappears once realistic SWAP inflation is included at fixed per-gate noise? *Basis:* $p=2$ used 60 CX assuming all-to-all connectivity; a physical device would run significantly deeper.